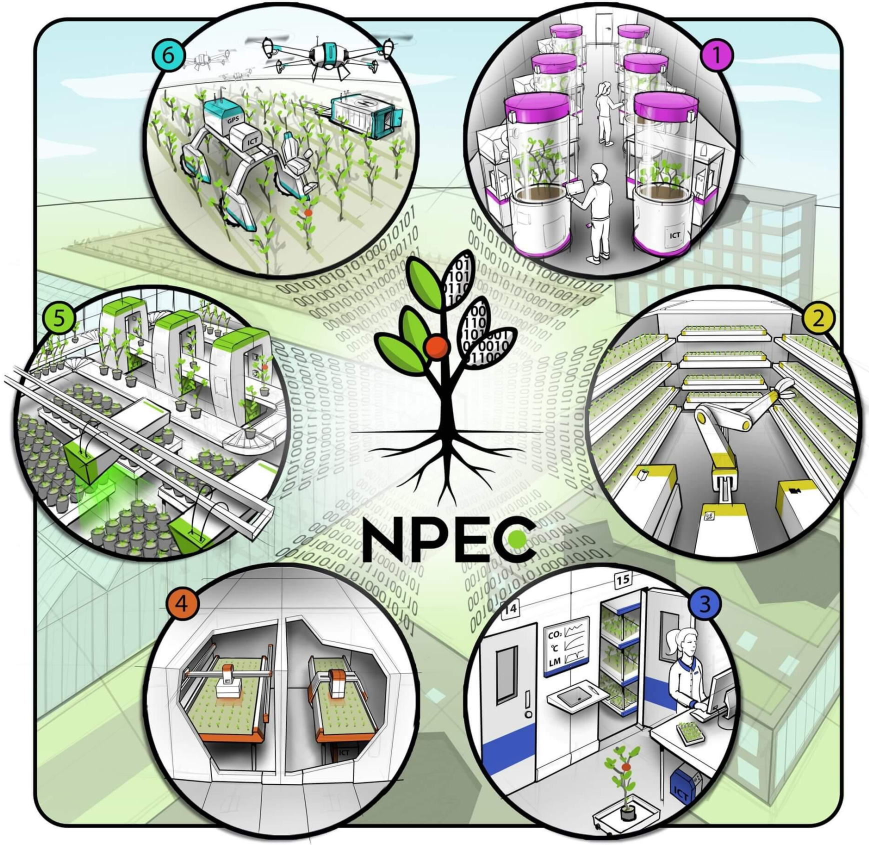
Plant Inoculation

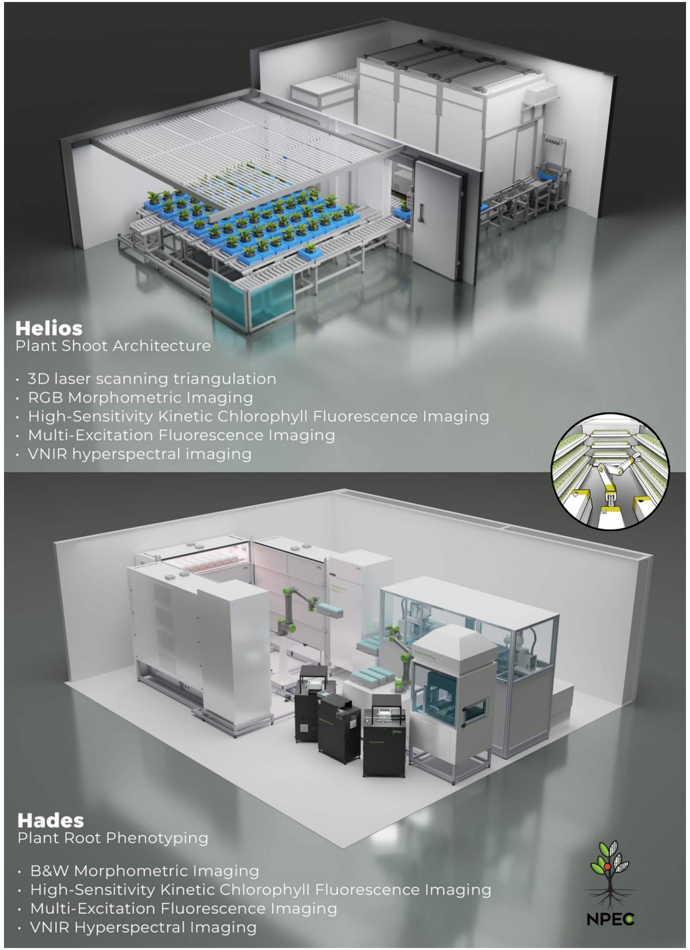
By: Kade Widler

NPEC

When Biology Evolves Faster Than We Do

NPEC: Plant Phenotyping





Problem Statement

Time Compression

- **Current State:**
  - 10-15 year breeding cycle
  - 0.7-1 selection cycles per decade
- **With Pipeline:**
  - 2-3 year breeding cycle
  - 3-5 selection cycles per decade
- **Impact:** Bring improved varieties to market 7-12 years earlier (KWS)

Future Impact

By 2050, we need crops that are:

- 25% more productive on 25% less arable land
- **Resilient to climate extremes** (drought, heat, flooding)
- Nutrient-efficient for sustainable farming

Root architecture is the key: Roots determine water/nutrient access, drought tolerance, and soil health

Current breeding pace is insufficient: At 10-15 years per variety, we can't keep pace with climate change (foundationfar)

Hades Pipeline Overview

Petri Dish Image

CNN Outputted Root Mask

Post-Processing + RSA

Communication with Hades

Accelerated Phenotyping

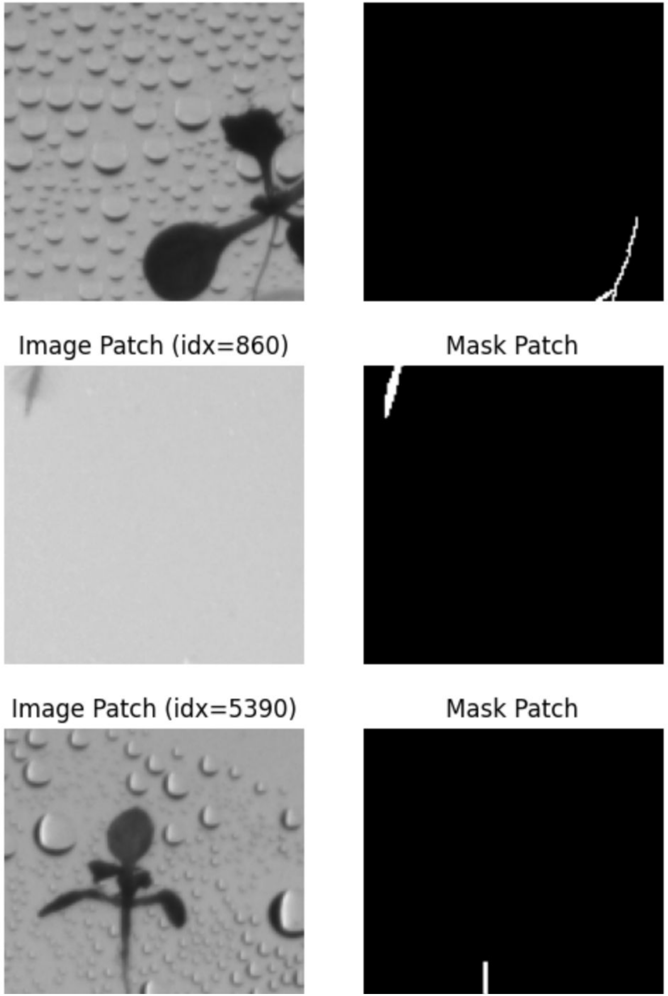
Automated Vision-Guided Robotic System

Business Impact

- Traditional breeding methods produce only 1-2 generations per year
- A single breeding cycle costs **$500K-2M+ annually** (CGIAR)
- Manual phenotyping is one of the highest cost centers in breeding programs (KALRO)

Proposed Solution

- **Speed:** Processes hundreds of plants per week (vs. dozens manually)
- **Precision:** Sub-millimeter accuracy for RSA measurements and targeted root-tip interventions
- **Consistency:** Eliminates human measurement variability

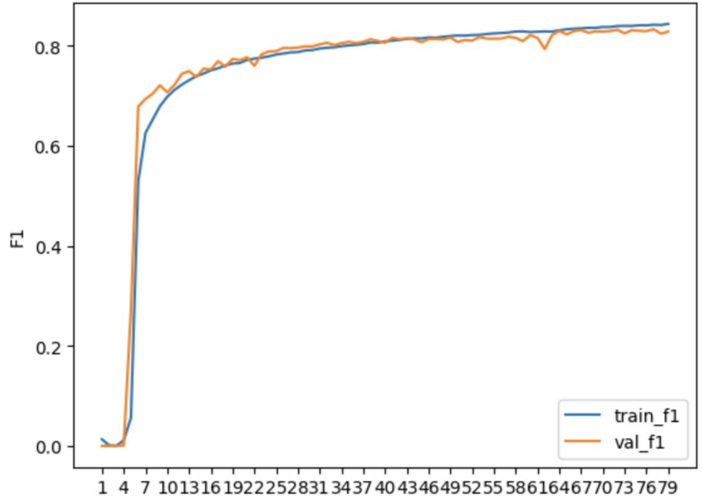


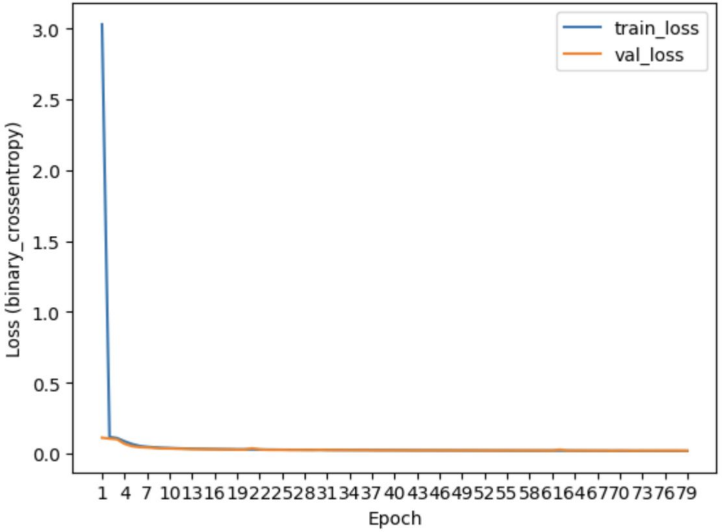
Dataset Creation

- ❑ Patch Size: 128
- ❑ Margin: 100
- ❑ Horizontal Augmentation
- ❑ Train: 9179 Test: 925

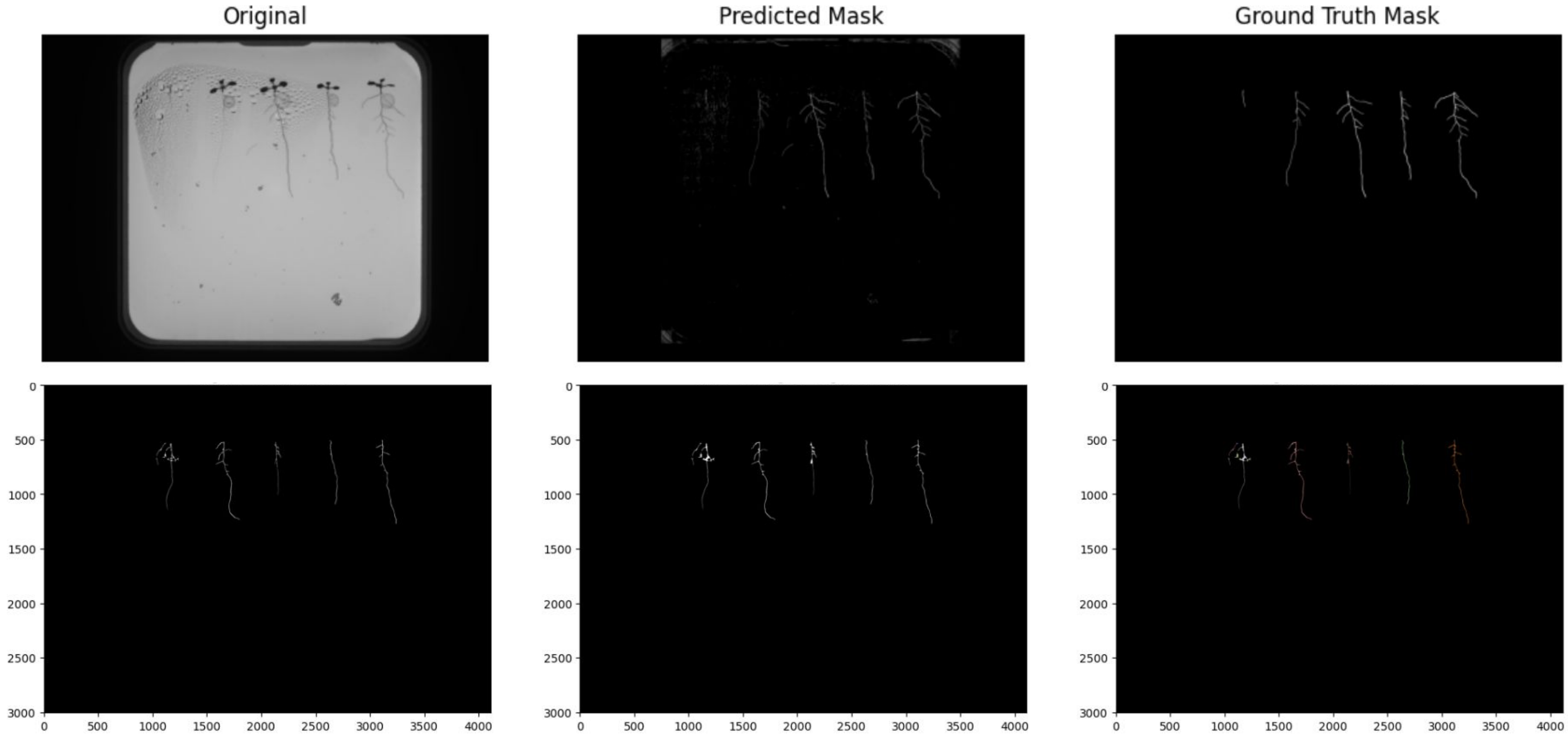
Model Training U-Net

- Basic U-Net architecture
- Image shape: (128, 128, 3)
- Binary Cross Entropy
- Best validation loss: 0.02
- Best validation f1: 0.83





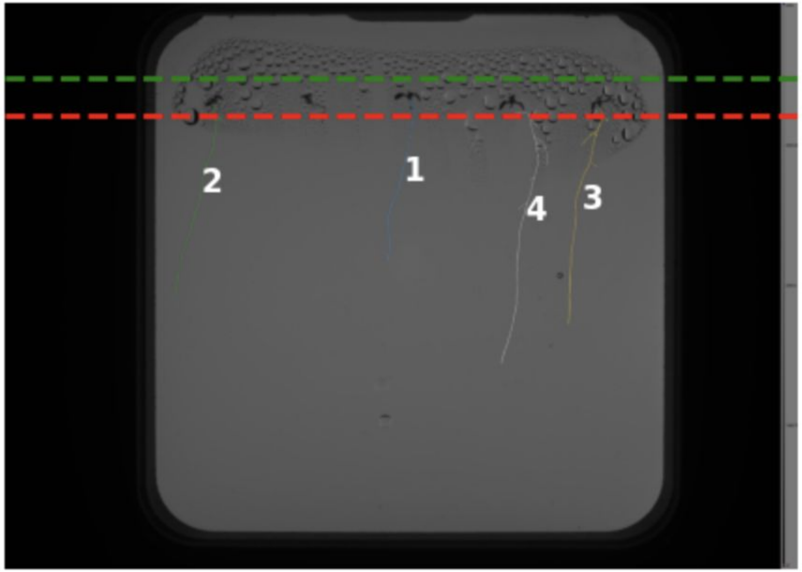
Inference, Cleaning, Segmentation, RSA –> Output Coordinates



Output Production

- ❏ Focused Noise Removal
- ❏ Shape and location
- ❏ Connected Components
- ❏ Learned positions
- ❏ Lowest Y Extraction

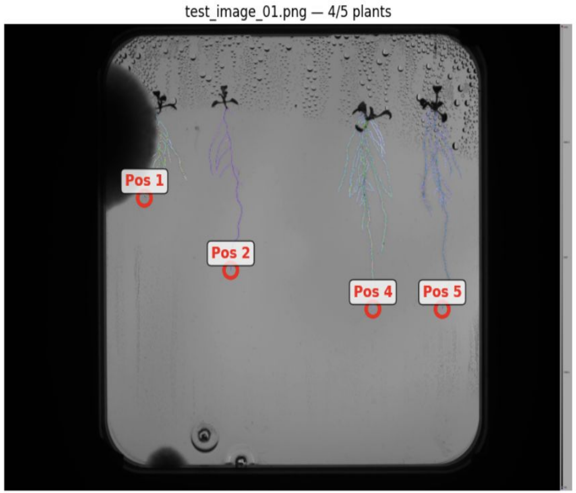
Segmented Plants: 4

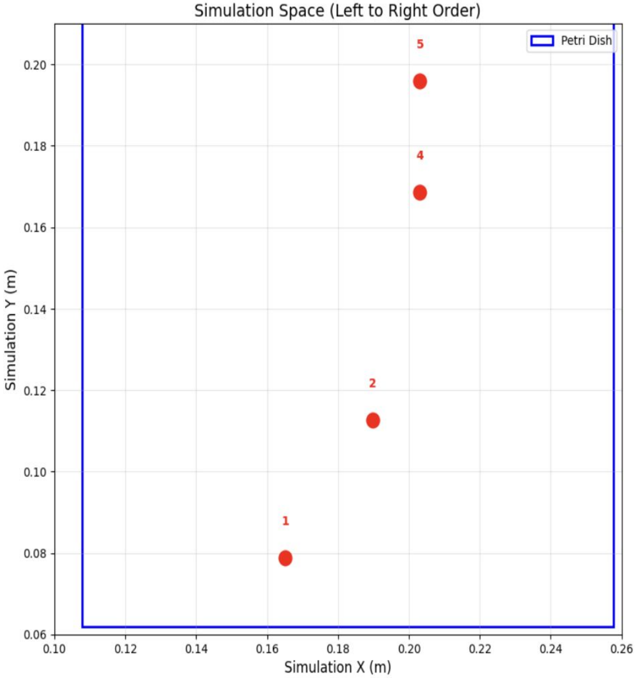


1665, 0, 1377, 2098, 1807

Pixels to MM

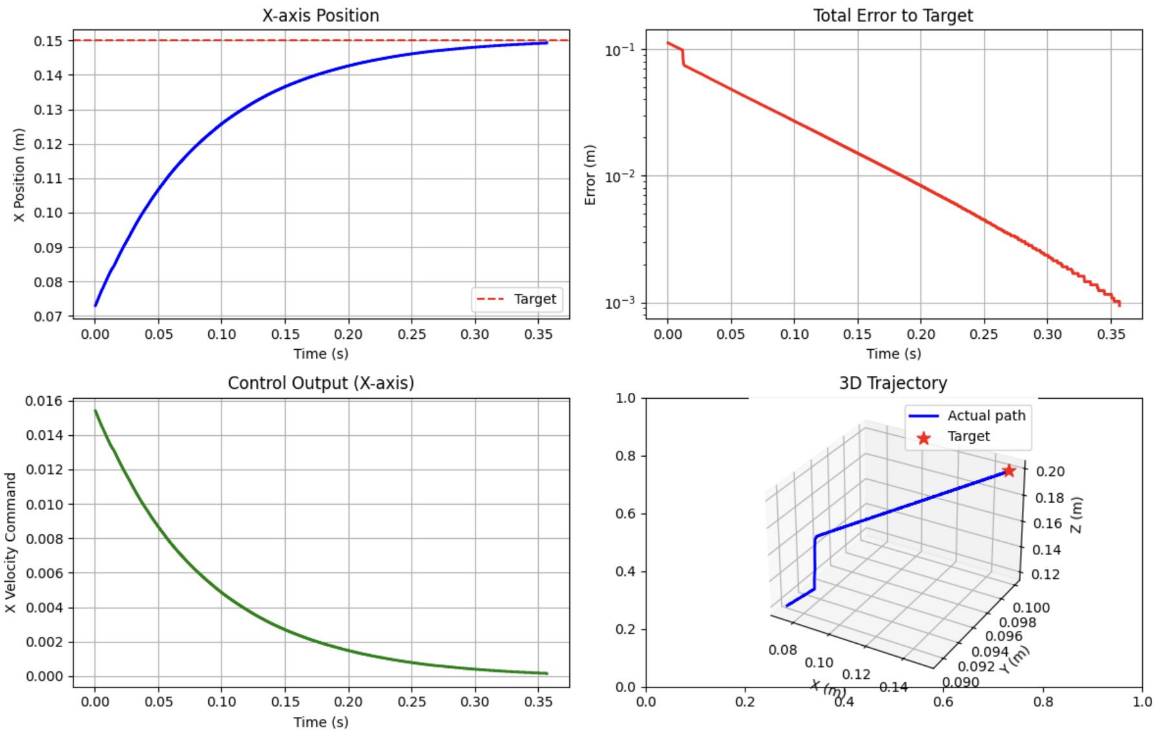
- Camera View
- 90 counterclockwise rotation
- X to Y Inverse





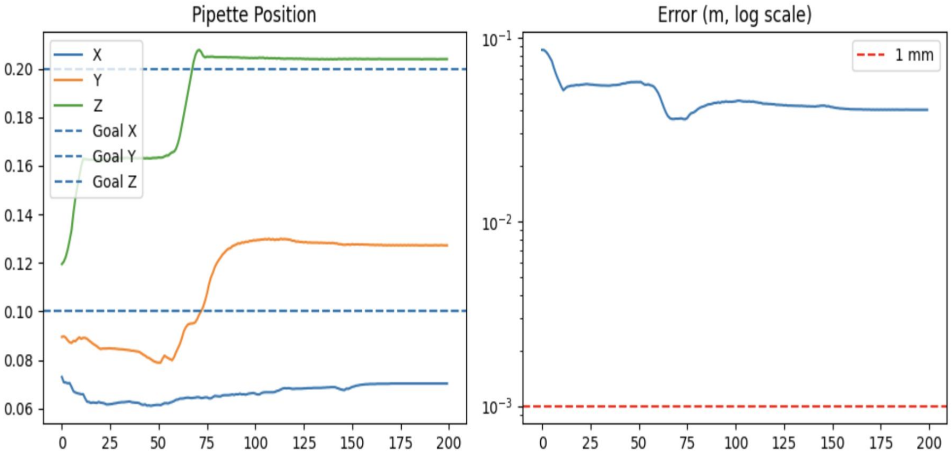
PID Controller

- Low Proportional: 0.10
- Derivative: 0.15
- Integral: 0.0006
- Grid Search




```
Final position: [0.1493, 0.0998, 0.1994]
Target position: [0.1500, 0.1000, 0.2000]
Final error: 0.94 mm
Time taken: 0.36 s
Oscillating: NO
```

Reinforcement Learning



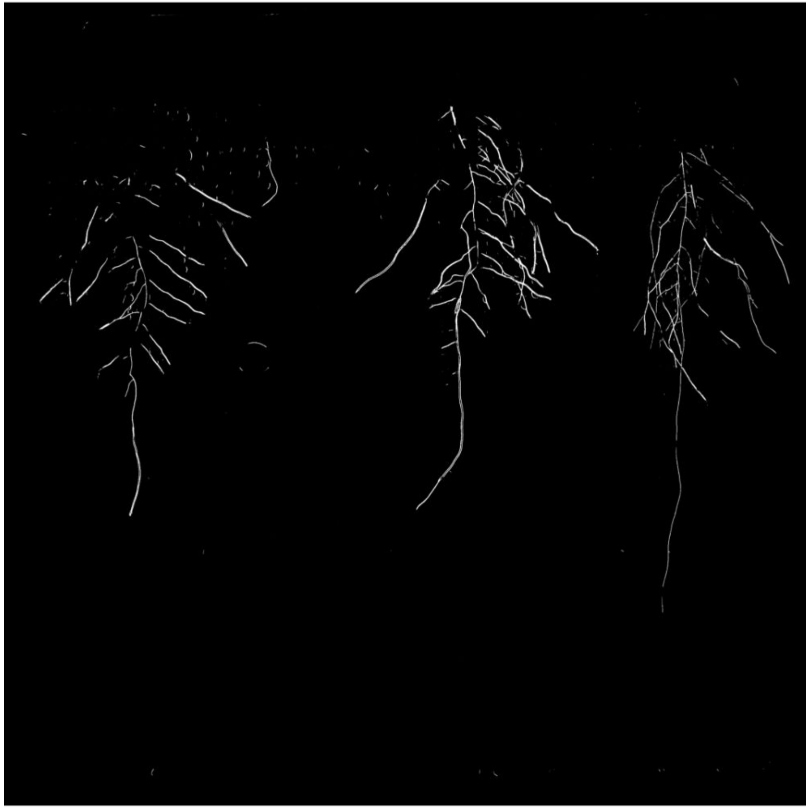
- ❑ PPO and SAC
  - ❑ Actions (probability distribution)
  - ❑ Reducing Environment Exploration
- ❑ 5 Million Timesteps
- ❑ Reward Shaping
- ❑ Progress Bonus

Integration Comparison RL vs PID

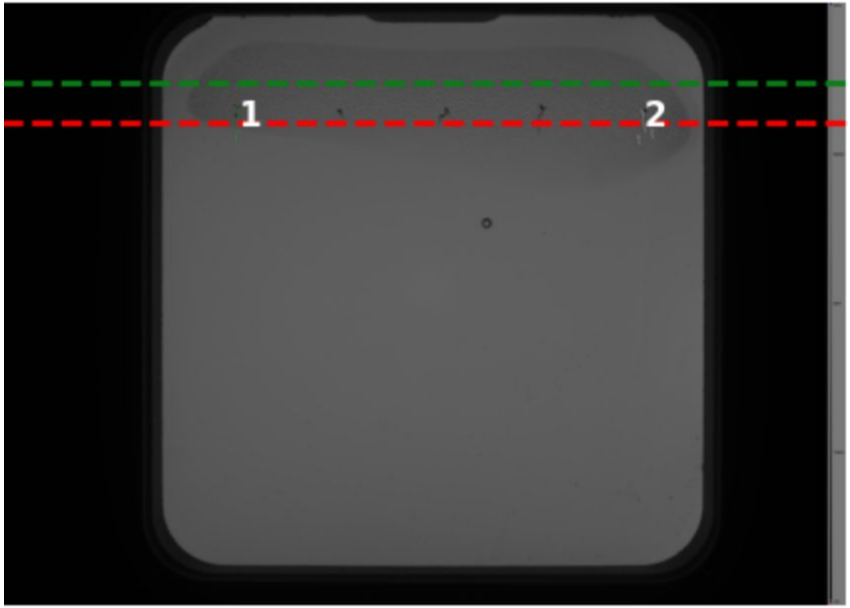
Metric	PID Controller	RL Controller
Success rate (≤1mm)	100%	0%
Mean error	0.85 mm	2.83 mm
Std error	0.12 mm	0.23 mm
Max error	1.25 mm	3.53 mm
Execution time per plant	0.05 sec	0.3 sec

Error Analysis: Dataset Iteration

My initial pipeline submission on Kaggle earned a **27% sMAPE** score. Due to the output of my model I had to implement strong post-processing (**min_size=150**) which removed noise, but also instances of smaller plants. To improve upon this I increased dataset size (added 2025 data + random **50% horizontal augmentation** + 15% empty patches). Data set size increased from **9,789 to 14,576** patches removing most noise from generated mask. I implemented only a **small closing kernel** to remove remaining noise and my performance increased to **17% sMAPE**.



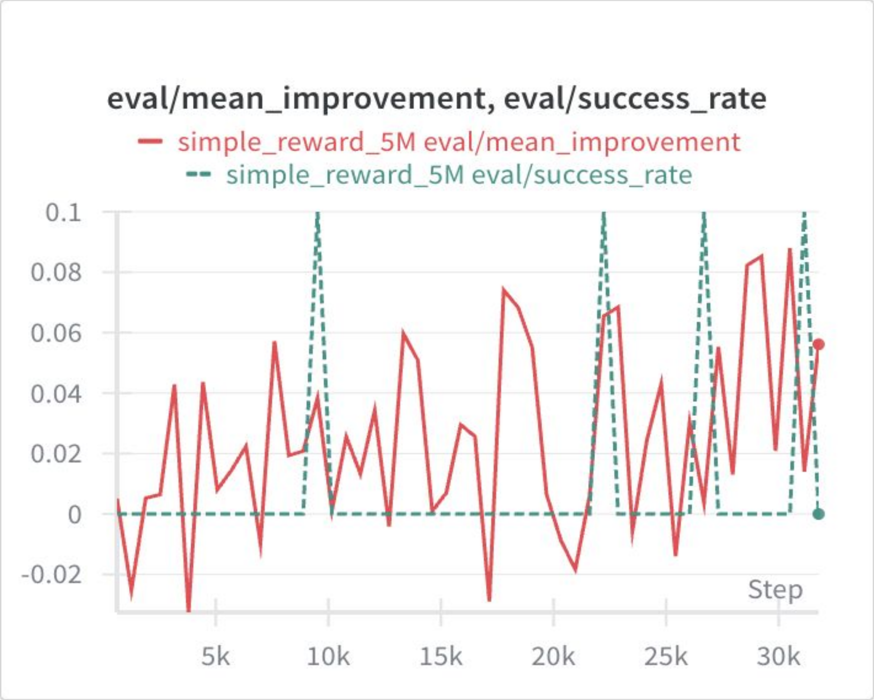
Error Analysis: Start Point Segmentation



I noticed my segmentation model was producing **false positive predictions** of remaining noise. Creating **hard boundaries** was not a robust method to sort the 5 plant predictions output order, as plants appeared in similar (but not identical) regions across images and roots extended and curved across x-boundaries. To solve this I introduced **X&Y-band filtering** and learned clustering on seed location. These methods were more robust because the seed locations were uniform across petri dishes. The presence of **plants detected increased from 72% to 89%**.

Error Analysis: RL training

Hard to know if model was learning due to **complex kinematics** and **low threshold** even with 80% improvement over epochs there was still 20% success. I switched to **SAC**, made the **simple reward (-distance)**, and introduced **iterative thresholding**. With a threshold of 1 cm: 60% improvement and success increased to 40%.



Assumptions: Environmental & Equipment Constraints

Single species pipeline:

- ❏ This system was developed and validated exclusively for arabidopsis thaliana plants in standardized petri dishes
- ❏ Pipeline assumes consistent plant morphology and growth patterns
- ❏ **Not tested** across different plant species or varieties

Controlled laboratory environment:

- ❏ **Assumes consistent lighting conditions, petri dish positioning, and camera angles**
- ❏ Has not been validated in production greenhouse or field environments
- ❏ No testing with environmental variations (temperature, humidity effects on condensation)

Hardware-specific calibration:

- ❏ PID controller parameters optimized only in PyBullet simulation environment
- ❏ Real-world implementation risk: Faster movement speeds could introduce vibrations/table shaking that affect precision
- ❏ **No validation** on physical OT-2 hardware with real mechanical constraints

Assumptions: Biological & Processing

Uniform seed positioning:

- Pipeline assumes seeds are placed in consistent spatial regions within petri dishes
- **Y-band filtering** and clustering depend on this spatial consistency
- Deviation from expected seed positions may cause detection failures

Complete inoculation protocol:

- All detected plants are inoculated without seed/germination verification
- **Assumes all visible root systems represent viable, germinated plants**
- No logic to skip non-viable or contaminated specimens

Early-stage plant bias:

- Training dataset predominantly consists of young plants (**early lifecycle stages**)
- Model performance on mature plants with complex root architectures is unvalidated

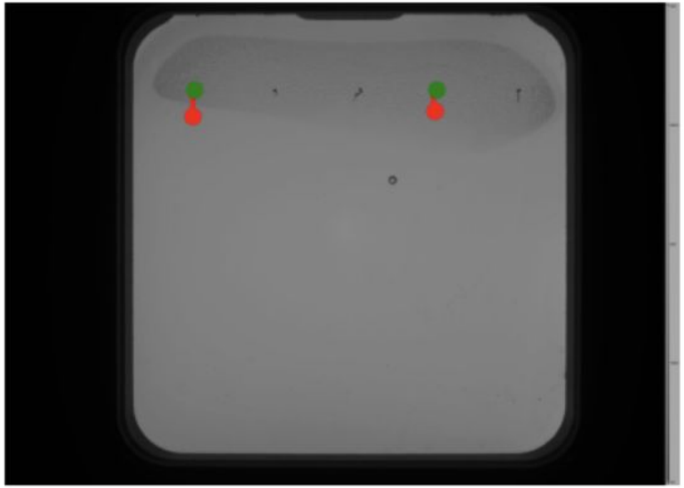
Limitations

1. Small Plants not Identified

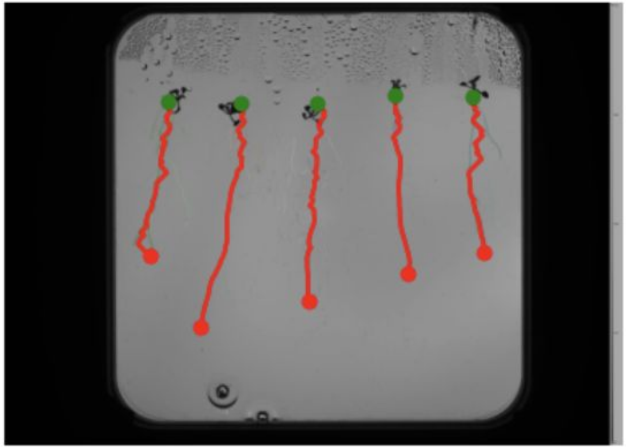
2. Incorrect Measurement Tracking

3. Root vs Shoot

test_image_05.png
Plants: 2



test_image_17.png
Plants: 5



Next Steps: Year One

Short-term (< 6 months):

Real-world validation campaign:

- ❑ **Shoot model**
- ❑ **Updated post-processing**
- ❑ **Deploy pipeline on physical OT-2 hardware**

Dataset enrichment:

- ❑ Collect and annotate images of **mature plants**
- ❑ **Include edge cases**: condensation, dish contamination, plant overlap
- ❑ **Retrain model** with expanded dataset for improved robustness

Medium-term (6-12 months):

Multi-species generalization:

- ❑ **Extend to additional crop species** (wheat, rice, legumes)
- ❑ Test on different petri dish formats and growth media
- ❑ Validate with **diseased/stressed plant** specimens

Comprehensive RSA phenotyping

(multi-task segmentation model):

- ❑ Lateral root count and density
- ❑ Shoot width and biomass estimation
- ❑ Root branching angles and architecture complexity
- ❑ Seed germination status detection

Next Steps: Long Term + Possibilities

Long-term (1-2 years):

Longitudinal tracking system:

- ❑ Implement plant identity tracking across multiple imaging sessions
- ❑ Build temporal growth models to analyze:
  - ❑ **Root elongation rates over time**
  - ❑ **Developmental stage transitions**
  - ❑ **Treatment response dynamics**

Production-scale deployment:

- ❑ Integration with automated **greenhouse conveyor systems**
- ❑ Multi-robot coordination for parallel processing
- ❑ Cloud-based data pipeline for real-time phenotyping analytics

Business Potential:

- ❑ Phenotyping Service Platform
- ❑ Turnkey System Sales
- ❑ Data Analytics

Global Impact Opportunities:

- ❑ **Climate-Resilient Crop Development**
- ❑ **Developing World Deployment**
- ❑ **Rapid Pandemic Response**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